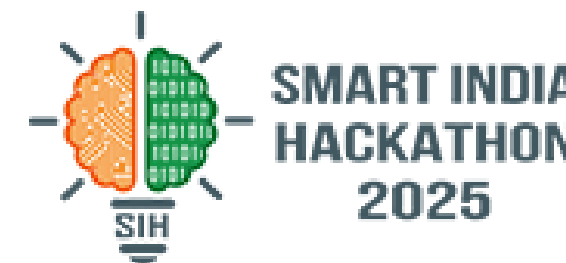




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SMART INDIA HACKATHON 2025



TITLE PAGE

Problem Statement Title -

**TRANSFORMER BASED END-TO-END WEB
APPLICATION FIREWALL (WAF) PIPELINE**

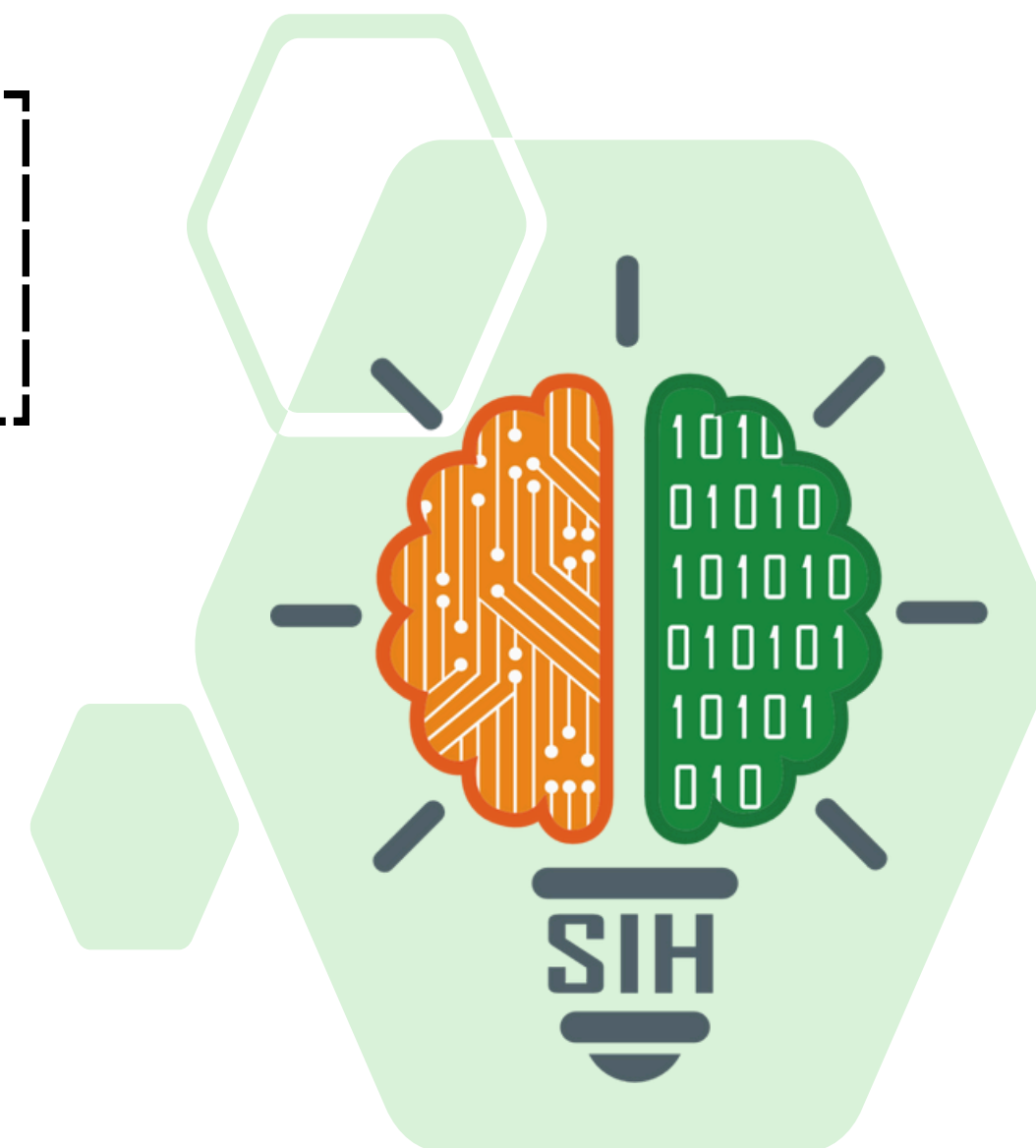
Problem Statement ID - SIH25172

Theme - Blockchain & Cybersecurity

PS Category - Software

Team ID - 83470

Team Name - Team M3OW (MACHINES, MATHEMATICS, AND MODELS OF WONDER)



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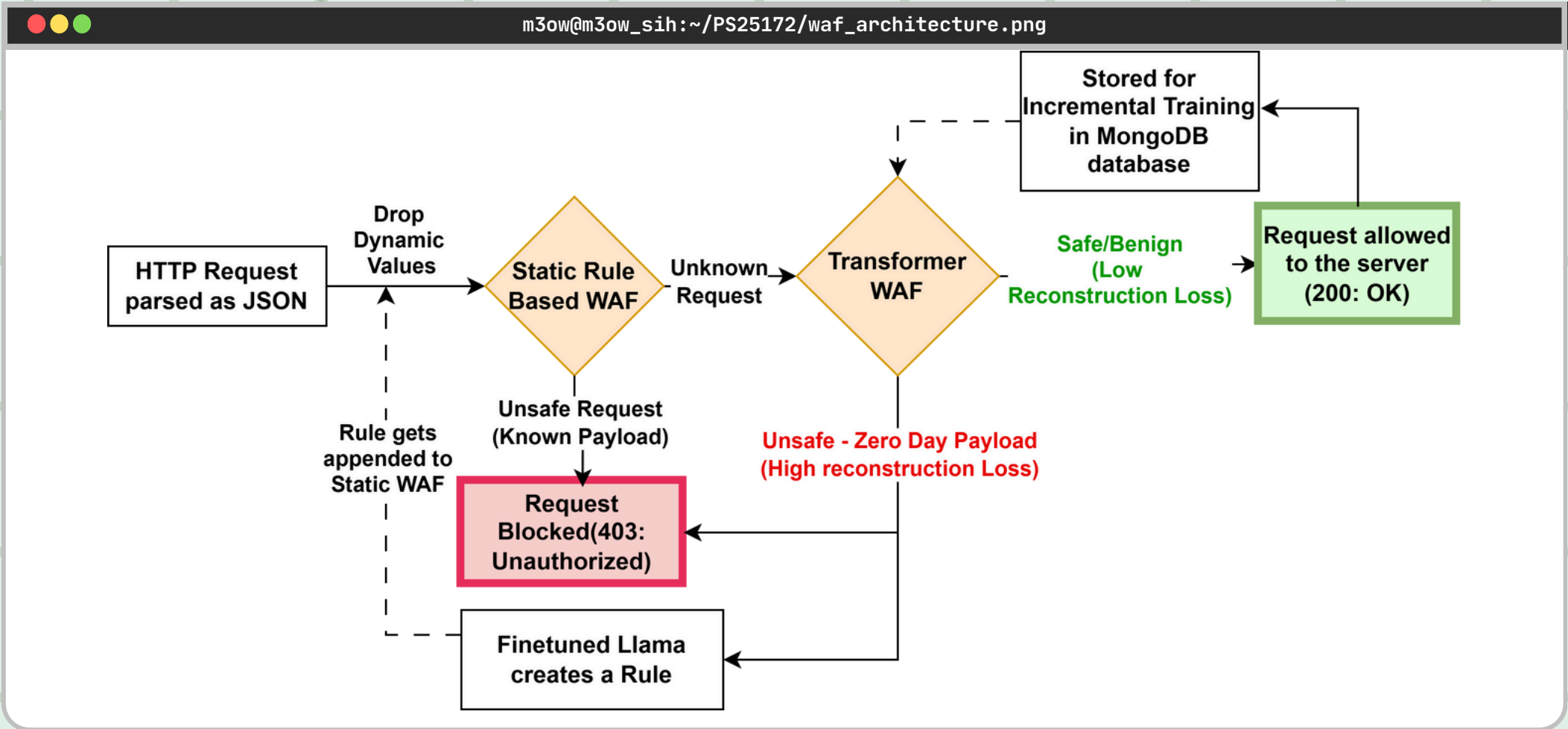
PROPOSED SOLUTION

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We propose a hybrid, self-hardening Web Application Firewall that integrates a static rule-based engine with a *Transformer-based model for zero-day threat detection*. All incoming HTTP requests are first evaluated by the static WAF. Requests matching known signatures are blocked, while unknown traffic is passed to a **fine-tuned DistilBERT model** that scores them based on **reconstruction loss** and a **separate Contrastive Classification head**. This is performed asynchronously to ensure **zero user latency**. The system features a dual-feedback loop:

1. Anomalous requests with high reconstruction loss are used by a **fine-tuned Llama model** to automatically **generate and append new signatures to the static rule set**.
2. While benign requests with **low loss** are stored for future **incremental model training**.

This architecture allows both the static and dynamic detection layers to continuously learn and harden against emerging threats.



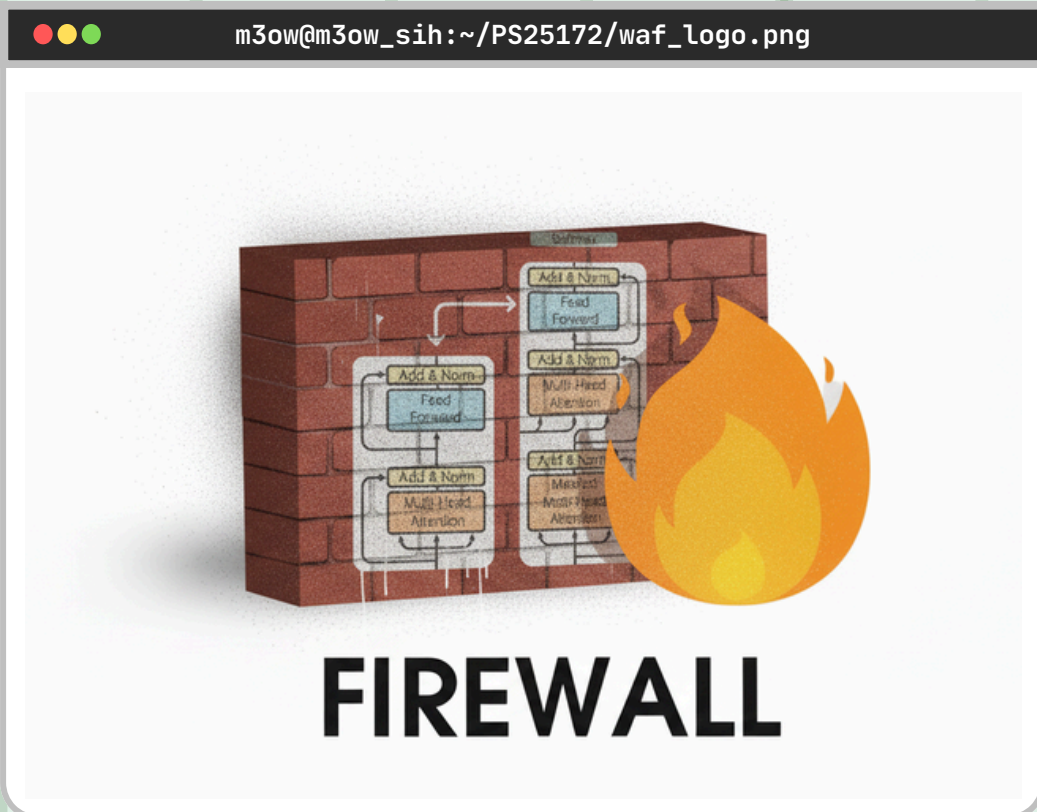
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FEATURES

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- Ensemble Methods - Contrastive Learning, Masked Language Modelling (MLM) & Statistical Analysis.
- Red Team - Blue Team Post-training based strengthening of WAF.
- Live Tracking of logs.
- Knowledge Distillation and Adapter usage for strengthening via incremental training
- Easy Integration with existing infrastructure.
- Near zero False Positives



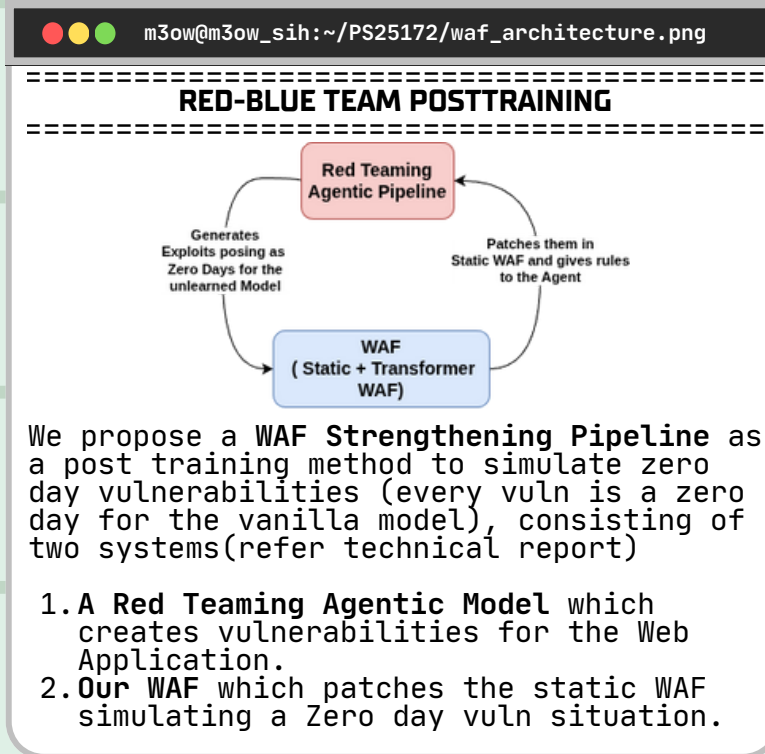
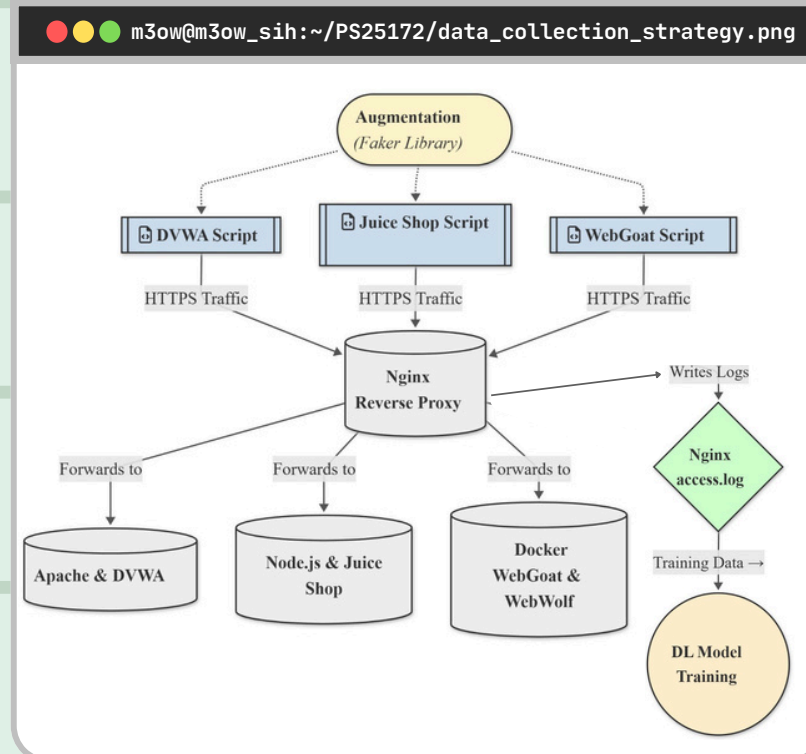
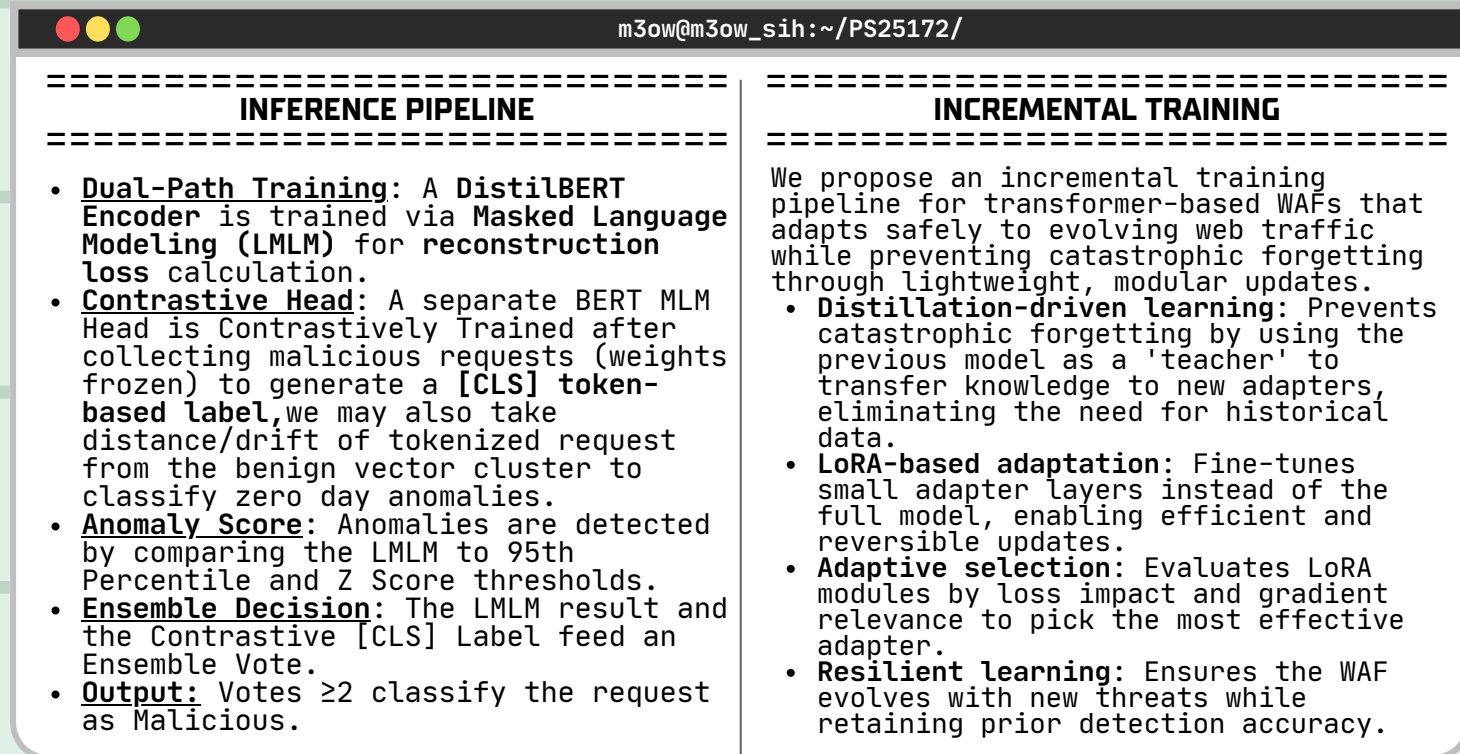
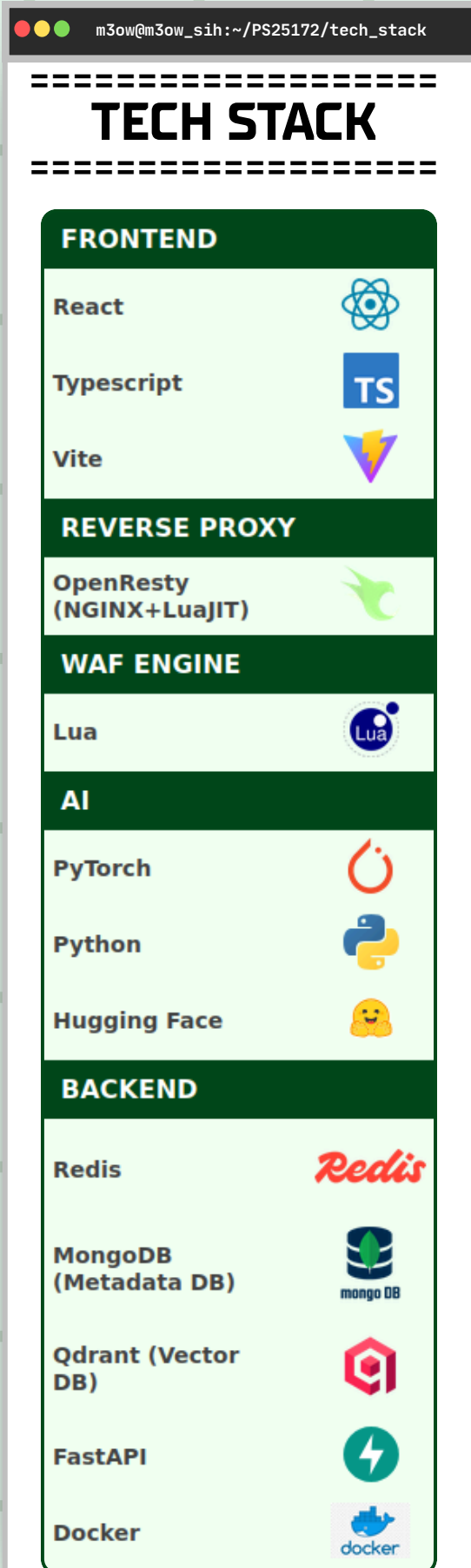
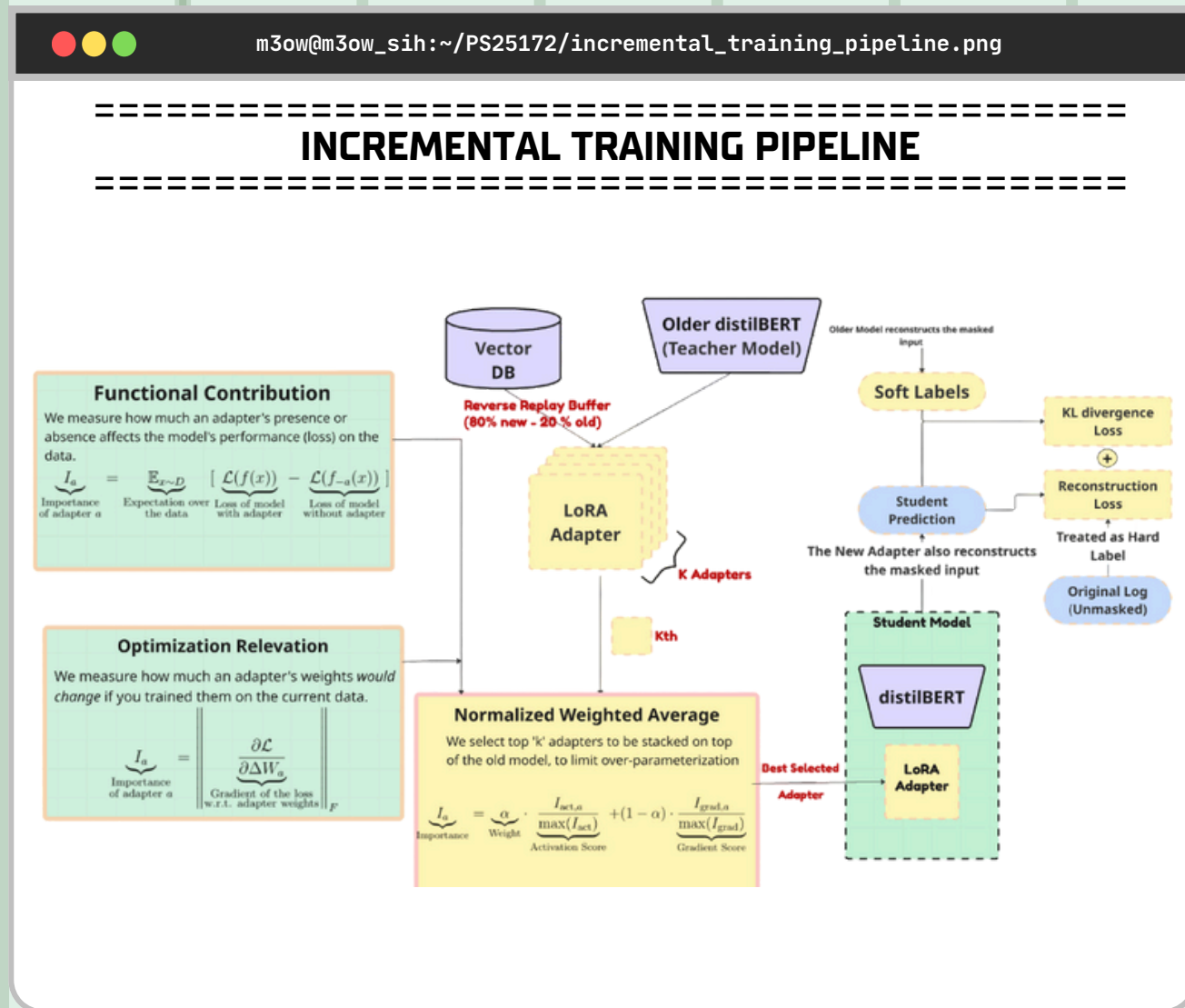
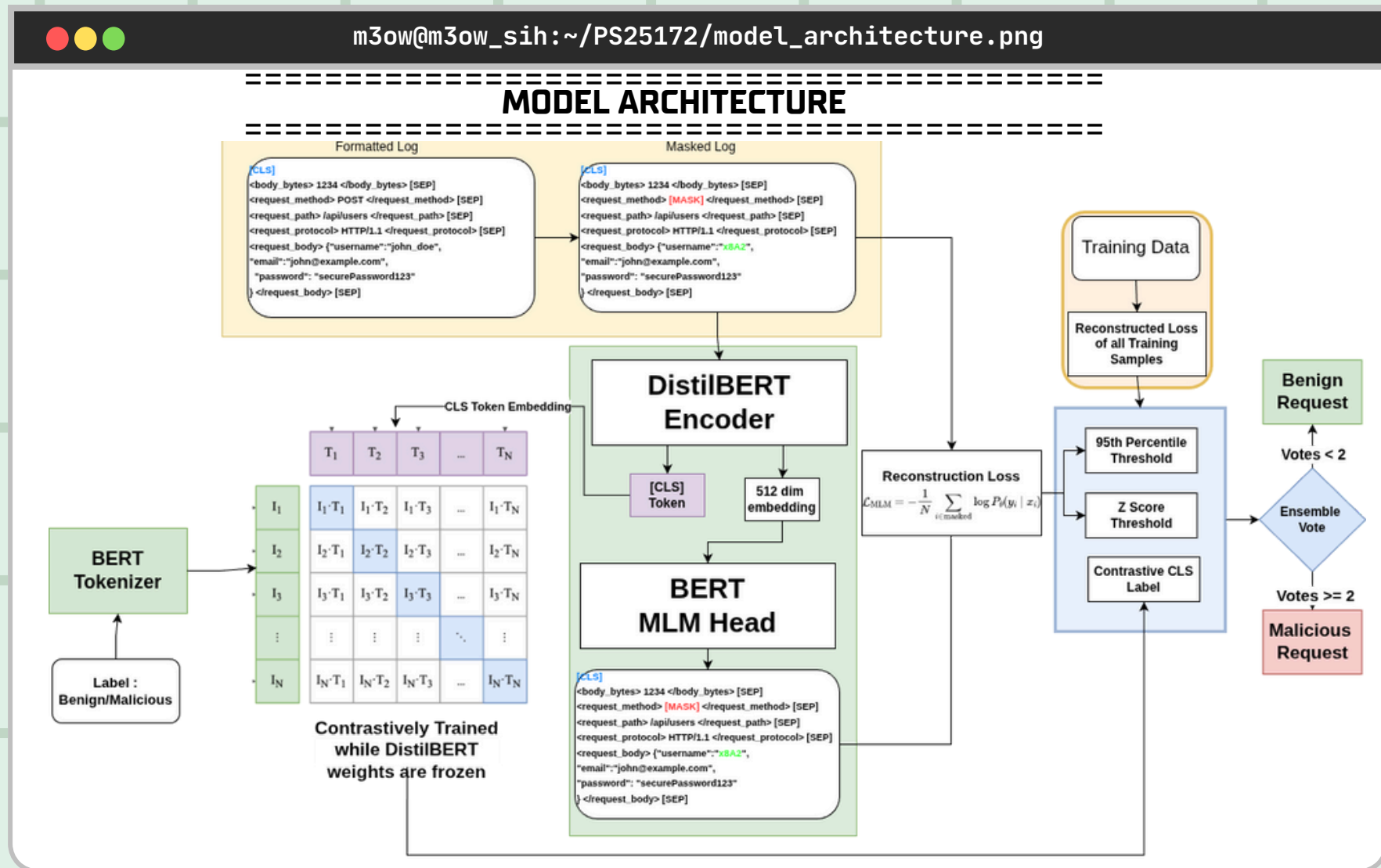
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UNIQUE SELLING PROPOSITION

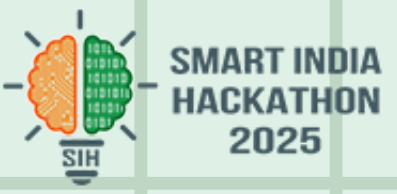
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- Toggle between Static + Transformer or Transformer-only modes.
- Optional Static WAF Layer provides: **WAF ARCHITECTURE**
Reduced inference latency for common exploits.
Self-healing capability by auto-patching rules via LLM-generated updates.
- Ensemble of Statistical and Deep Learning models ensures minimal false positives and negatives.
- Real-time Tracking with Human-in-the-Loop supervision enhances reliability.
- Incremental Learning achieved through synergistic operation of Knowledge Distillation (KD) and LoRA-like adapters for adaptive model refinement.





FEASIBILITY & VIABILITY



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ACCURACY METRICS

Anomaly Catch
Rate (Recall)

93.78%
(2280/2431)

High Zero-Day Coverage: Our model effectively detects novel, never-before-seen injection payloads and zero-day attacks, which traditional rule-sets miss.

Missed Attack
Rate (FNR)

6.17%
(151/2431)

Acceptable Breach Risk: Only 6.17% of malicious requests bypass the system. The continuous learning loop is designed to reduce this further incrementally.



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FEASIBILITY & VIABILITY

The model was easily deployed for batched inference (batch size = 4) on an RTX 4080 (24 GB VRAM) and performed efficiently at prototype scale. Scaling to multi-GPU clusters or cloud accelerators (e.g., A100/H100) is straightforward, offering minimal latency compared to GPU-intensive LLM proxy setups. Lightweight LoRA adapters cut fine-tuning costs to ~20% of full training. The system is both technically and economically viable, enabling continuous learning and real-time adaptation with far lower overhead than static, rule-based defenses.

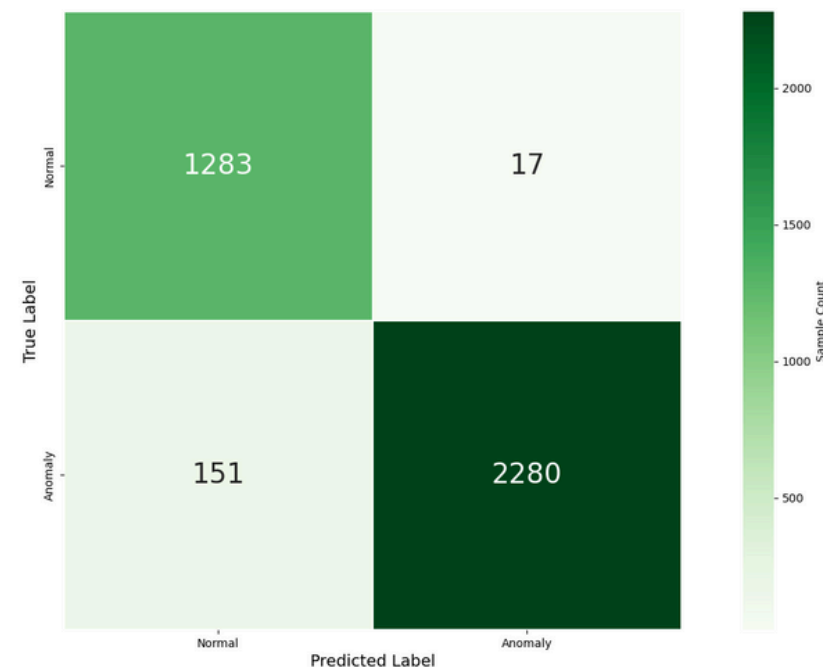
Deployed as a cloud-based Firewall-as-a-Service (FWaaS), the system runs on GPU clusters behind a load-balanced ingress. Client traffic is normalized, analyzed, and filtered in real time based on model confidence and policy rules. The design supports multi-tenancy with isolated model namespaces and usage-based billing tied to website traffic.



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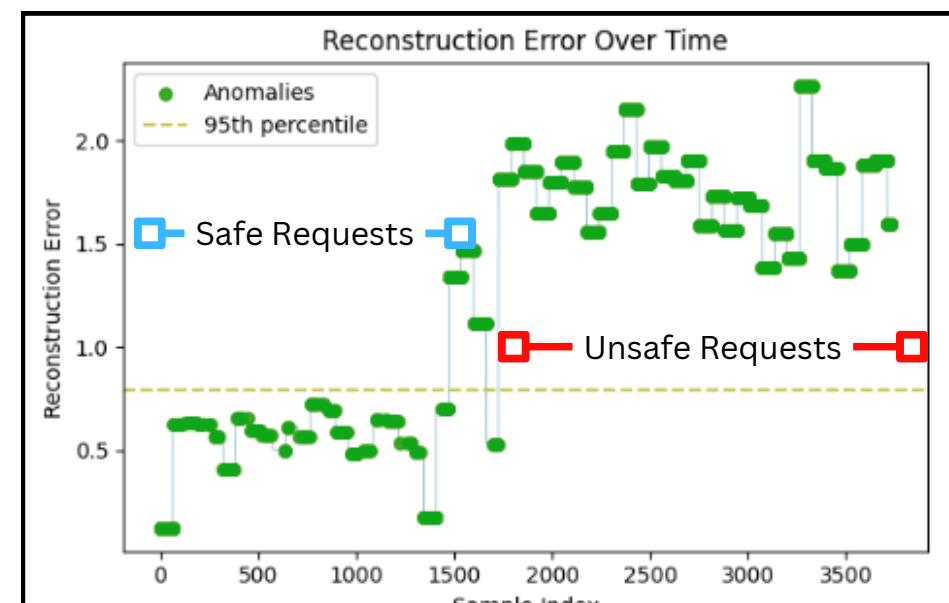
CONFUSION MATRIX

Confusion Matrix (Ensemble)



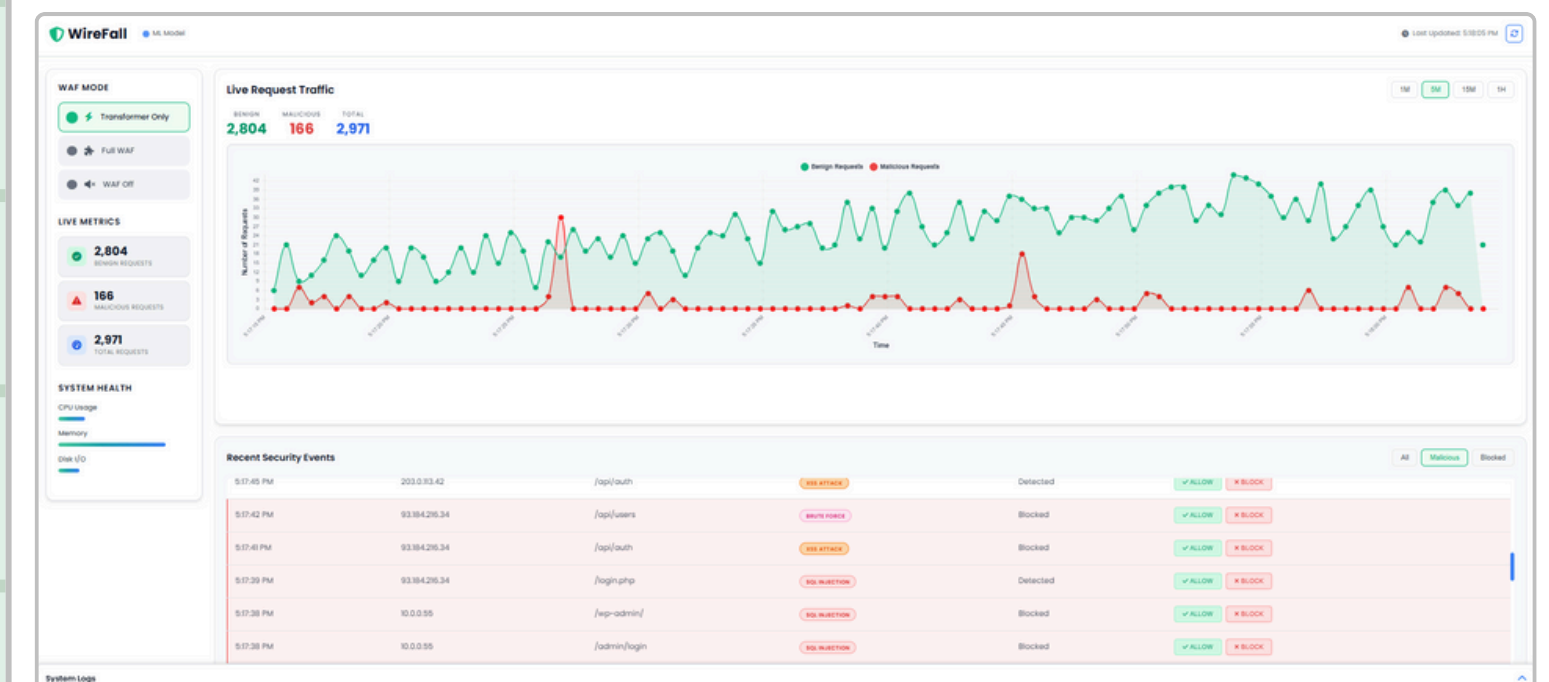
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ANOMALY SCORES OVER TEST SET



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WEB DASHBOARD





IMPACTS AND BENEFITS



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ADOPTION SECTORS & POTENTIAL COLLABORATORS

Critical Information Infrastructure (CII)	ISRO (Space Systems & Telemetry), DRDO (Defense R&D), NCIIPC (Protection Centre), Indian Military (Cyber Command), etc.
Financial & Regulatory Compliance	NPCI (UPI, Payment Gateways), RBI-Regulated Banks (SBI, HDFC Bank, Axis Bank), Fintech Platforms, etc.
High-Volume E-Services	Government Portals (MyGov, Digital India projects), Large E-commerce Platforms (Flipkart, Amazon India), Telecom (Jio, Airtel), etc.
Technology & AIOps Partners	TCS, Wipro, Infosys, HCLTech, Domestic Cloud Providers, etc.

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FUTURE SCOPE

- Addition of **Energy based Models** into the WAF Ensemble Pipeline (high energy = anomaly, low energy = benign)
- Usage of **Log Based Attention Methods** for Encoder only Transformer, to accommodate the semantic and structural value of a Log
- Faster Inference Time with techniques like **Flash Attention**.
- Experimentation and Research for **better incremental training methods**, suited to high traffic domain like WAF.
- Application to other domains like **Network Security**.
- **Agentic Pipelines for Red Teaming Agents** like a **LLM Based Fuzzers** and **Pentesters** to improve the strength of the firewall, involving a **RL based training**

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OPERATIONAL & BUSINESS IMPACT

False Positive Rate (FPR)	1.30% (17 / 1300)	Current Alert Challenge: The current threshold yields a high FPR. Our rule based pre-check and the proposed LLM-driven rule feedback will be used to automatically tune this threshold down to <1% in deployment.
Performance	Non-Blocking Asynchronous Design	Zero Latency: Decoupling the deep learning model (via Redis streams) ensures concurrent scanning for zero latency. This architecture maintains the high throughput required by national platforms (e.g. UPI, Aadhaar services).
Maintainability	LoRA Incremental Fine-Tuning	Adaptive & Low-Cost: Eliminates manual WAF rule maintenance. The system automatically and incrementally updates its model knowledge, drastically reducing operational cost.
Reduces Financial Risk	High F1-Score (93.60%)	DPDP & Regulatory Shield: Provides superior protection against breaches and data exfiltration. This reduces financial risks related to regulatory penalties/fines under the DPDP Act and minimizing system downtime costs.



RESEARCH AND REFERENCES

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LINKS

- Youtube Playlist:
<https://tinyurl.com/m3ow-wirefall>
- Github :
<https://github.com/Team-M3OW/WireFall>
- Technical Analysis Report:
https://wirefall.hacker101.xyz/technical_report
- Website:
<https://wirefall.hacker101.xyz/>

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REFERENCES

- DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter
- Learning Transferable Visual Models From Natural Language Supervision (CLIP)
- LLaMA: Open and Efficient Foundation Language Models
- LoRA: Low-Rank Adaptation of Large Language Models
- Data-Distill-Net: A Data Distillation Approach Tailored for Reply-based Continual Learning
- Attention Is All You Need
- Continual Learning with Pre-Trained Models: A Survey
- A Simple Framework for Contrastive Learning of Visual Representations
- LAnoBERT: System Log Anomaly Detection based on BERT Masked Language Model